

2.8 Product Rule: problem 9

9. $d(t) = 3\sqrt{t} \ln t$

Rewrite the root as a power: $\sqrt{t} = t^{1/2}$

$$d(t) = 3t^{1/2} \ln t$$

$$(uv)' = u'v + uv'$$

Let:

- $u = 3t^{1/2} \rightarrow u' = \frac{3}{2}t^{-1/2}$
- $v = \ln t \rightarrow v' = \frac{1}{t}$
- $u' = 3 \cdot \frac{1}{2} t^{\frac{1}{2}-1} = \frac{3}{2} t^{-1/2}$ (power rule: bring down $\frac{1}{2}$, subtract 1)

So:

$$d'(t) = u'v + uv'$$

$$d'(t) = \left(\frac{3}{2}t^{-1/2}\right)(\ln t) + \left(3t^{1/2}\right)\left(\frac{1}{t}\right)$$

First term: $\frac{3}{2}t^{-1/2} \ln t = \frac{3 \ln t}{2\sqrt{t}}$

Second term: $3t^{1/2} \cdot t^{-1} = 3t^{1/2-1} = 3t^{-1/2} = \frac{3}{\sqrt{t}}$

$$d'(t) = \frac{3 \ln t}{2\sqrt{t}} + \frac{3}{\sqrt{t}}$$

Common denominator: $\frac{3}{\sqrt{t}} = \frac{6}{2\sqrt{t}}$

$$d'(t) = \frac{3 \ln t}{2\sqrt{t}} + \frac{6}{2\sqrt{t}}$$

$$d'(t) = \frac{3 \ln t + 6}{2\sqrt{t}} = \frac{3(\ln t + 2)}{2\sqrt{t}}$$